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21 April 1966

MILITARY SPECIFICATION
PROJECTILE, H.E., 4.2 INCH, M329A1
METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Projectile, H.E., 4.2 inch, M329A1, Metal Parts Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550 - Ammunition and Special Weapons, General Specification for

STANDARDS

Federal

Federal Test Method Standard No. 141 - Paint, Varnish, Lacquer and Related Materials, Methods of Inspection, Sampling and Testing.

Federal Test Method Standard No. 151 - Metals; Test Methods

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-109 - Quality Assurance Terms and Definitions

MIL-STD-1167 - Ammunition Data Card

MIL-STD-1168 - Lot Numbering of Ammunition

MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC 1315

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DRAWINGS

Munitions Command
F-10523981 - Projectile, H.E., 4.2 inch, M329A1, Metal Parts Assembly
F-8848553 - Pallet
IEL-10523981 - Inspection Equipment List for Projectile, H.E., 4.2 inch, M329A1, Metal Parts Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA - B46.1 Surface Texture, Surface Roughness, Waviness and Lay.

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications, and the following:

3.2 Heat Treatment or Stress Relief.-Heat treatment or stress relief applied shall be uniform throughout each lot of components and shall be so conducted as to produce in each component the mechanical properties specified on the applicable drawing.

3.2.1 Hot Forge - Heat Treatment Method.-Prior to finish machining projectile bodies or bases manufactured from forgings shall be subjected to heat treatment to attain the required mechanical properties. The heat treatment shall consist of heating above the critical temperature, not to exceed 1600 degrees Fahrenheit (^{o}F), followed by an oil quench. After quenching, the forgings shall be tempered without delay at a temperature above 750° F. suitable to produce required mechanical properties. After removal from the tempering furnace, the forging shall be slowly cooled in air (forced air permitted) to a temperature of not more than 700° F., after which cooling may be accelarated by water spray.

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3.2.1.1 Bases (Alternative Design).-Bases may be heat treated by normalizing in lieu of quenching and tempering to obtain specified mechanical properties. The bases shall be heated to a temperature of 1600 °F., minimum, held at temperature for a minimum time of 30 minutes, and cooled in such a manner as to produce uniform mechanical properties throughout each component.

3.2.2 Cold Shaping and Hot Cup - Cold Draw Method.-Cold working shall be so accomplished as to produce in the projectile body or base the mechanical properties specified. The projectile body or base shall be stress relieved at a temperature of 700° F., minimum, following the final forming operation during which time plastic flow takes place. The minimum soak time at or above minimum temperature shall be established which will produce mechanical properties specified on the applicable drawing. The temperature and time duration of a process anneal prior to final forming operation shall be established by the ability to meet the required mechanical properties when formed by an approved sequence of operations.

3.2.3 Tubing.-Bodies (Alternative Design) manufactured from tubing which has complied with the mechanical properties specified on the applicable drawing shall be stress relieved after cold forming of the nose.

3.3 Mechanical Properties.-Mechanical properties as specified on the applicable component drawing shall apply throughout the component.

3.4 Quality of Joint.-The total area of inclusions, lack of adhesion, solder groove or other defects shall not exceed 25 percent of the total faying surfaces of the joint.

3.5 Functional

3.5.1 Excess Pressure.-There shall be no separation or loss of components in flight (nor shall there be evidence of failure of components due to firing stresses.

3.6 Workmanship.-The requirements for workmanship are as shown on the applicable drawing in the applicable specifications and the following:

3.6.1 Metal Defects.-All components shall be free from cracks, splits, cold shuts, pipe, porosity, inclusions or any similar defect.

3.6.2 Surface Finish.-The requirements for surface finish are as detailed on the applicable drawing. Each body cavity shall be free of scale, fins, deep seams, die scratches, pits or other surface discontinuities.

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3.6.3 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as specified on the applicable drawing, and shall be in accordance with the provisions of MIL-A-2550.

3.6.4 Painting.-Painting shall comply with the requirements of the applicable drawing. All paint shall be dry to the touch before testing and packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification.

3.6.5 Burr.-No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.6.6 Foreign Matter.-No part or assembly shall contain dirt, chips, grease, oil, rust, corrosion or other foreign matter. Particular attention shall be given to the charge cavity of the assembly, to insure that no grit, scale or other foreign matter remains therein after final cleaning.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance Terms and Definitions.-Reference shall be made to MIL-STD-109 to define Quality Assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 First Article Sample

4.2.1 Initial Production Sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 55 Metal Parts Assemblies as follows:

45 Metal Parts Assemblies Complete including Paint
5 Metal Parts Assemblies Complete except Paint
5 Sets Component Parts Unassembled and Unpainted

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4.2.1 (Cont'd)

Forty of the complete and painted metal parts assemblies shall be forwarded to a designated proving ground for ballistic testing. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing shall be the same as used in regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and Test.-After inspection and provisional acceptance at source, the sample will be inspected for all the requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial Production Sample Failure.-Failure of the sample to comply with the requirements of the drawings and specifications will result in sample disapproval.

4.3 Inspection Provisions

4.3.1 Lot

4.3.1.1 Submission of Product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Heat Treatment Lots.-Heat treatment lots shall contain body or base forgings from not more than one heat of steel that have been heat treated in the same heat treating equipment, at the same temperature, in one unchanged process. The heat treatment lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.3 Stress Relieved Lots.-Stress relieved lots shall contain cold worked bodies or bases from not more than one heat of steel that have been cold worked by the same process, utilizing the same tooling and equipment and stress relieved in the same furnance at the same temperature for the same length of time. The stress relieved lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.4 Tubing Lots.-Bodies manufactured from tubing shall be fabricated from tubing lots containing not more than one heat of steel. The bodies shall be cold nosed and stress relieved using the same equipment and procedures. Lots of bodies thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

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4.3.1.4.1 Adapters.-Adapters manufactured from tubing shall be fabricated from tubing lots containing not more than one heat of steel. Tubing lots thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.5 Bar Stock Lots.-Bar stock lots shall contain not more than one heat of steel. Bar stock lots thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.6 Miscellaneous Lots.-Small lots (such as bodies manufactured from abnormally small heats of steel or bodies and bases that have become separated from previously delivered lots, etc.) may at the discretion of the supplier be grouped into miscellaneous heat treatment, stress relieved or tubing lots as the case may be. Such lots shall be identified by a miscellaneous lot code number and processed in accordance with 4.3.1.2, 4.3.1.3 or 4.3.1.4 as applicable. The miscellaneous lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.7 Projectile Lots.-When it has been determined that a heat treated, stress relieved, tubing, bar stock or miscellaneous lots of bodies or bases have complied with the mechanical properties requirement of the applicable drawing, they may be regrouped for assembly into projectile lots, to which will be assigned a projectile lot number in accordance with MIL-STD-1168(MU). The projectile lot shall consist of projectile assemblies produced by one supplier under one contract in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision, under conditions of continuous production free from interruptions other than those due to the end of a shift, day or workweek. The projectile lot thus constituted shall be the basis for the selection of samples for the functional test hereinafter prescribed.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for MAJOR and MINOR defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. An AQL of 0.40 percent for each MAJOR defect and an AQL of 0.65 percent for each MINOR defect shall be used. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. When performing pressure tests, the item being tested will be observed for non-compliance during the entire time it is required to be pressurized. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter. The roughness comparison specimens prescribed by ASA - B46.1 shall be used as a basis for surface roughness determination.

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4.3.2.1 Adapter, Nose, Drawing D10534166, covering a detail of Drawing F-10523981

Defect		Method of Inspection
CRITICAL		
	None Defined	
MAJOR		
101.	Diameter of Brazing area	Gage
102.	Length of Brazing area	Gage
103.	Length from front to shoulder	Gage
104.	Metal defective	Visual
MINOR		
201.	Diameter of shoulder	Gage
202.	Presence and location of solder ring groove (if applicable)	Visual (see 6.2)
203.	Presence and location of staking notches	Visual (see 6.2)
204.	Radii or chamfers, missing or incorrect	Visual (see 6.2)
205.	Burr	Visual

4.3.2.2 Body, Drawing, D10534168 (Alternate Design) and Drawing F10534429 covering a detail of Drawing F-10523981

CRITICAL		
1.	Metal defective	Visual
MAJOR		
101.	Concentricity of charge cavity with bourrelets	Gage
102.	Diameter of brazing area, front	Gage
103.	Diameter of brazing area, rear (if applicable)	Gage
104.	Depth of brazing surface, at rear (if applicable)	Gage
105.	Diameter of charge cavity	Gage
106.	Variation in wall thickness, max.	Gage
MINOR		
201.	Diameter at front, outside	Gage
202.	Length to basic diameter	Gage
203.	Length to rear of forward bourrelet	Gage
204.	Length to front of rear bourrelet	Gage
205.	Length total	Gage
206.	Radii or chamfers, missing or incorrect	Visual (see 6.2)
207.	Surface finish, improper	Visual (see 6.2)
208.	Burr	Visual
209.	Foreign matter	Visual

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4.3.2.3 Base, Drawing D10534167 (Alternate Design) covering a detail of Drawing F10523981

Defect	Method of Inspection
CRITICAL	
1. Metal Defective	Gage
MAJOR	
101. Diameter of brazing area	Gage
102. Diameter of flange	Gage
103. Thickness of flange and disc seat	Gage
104. Length total	Gage
MINOR	
201. Presence and location of solder ring groove (if applicable)	Visual (see 6.2)
202. Surface finish, improper	Visual (see 6.2)
203. Radii or chamfers, missing or incorrect	Visual (see 6.2)
204. Burr	Visual

4.3.2.4 Metal Parts Assembly (Prior to Paint) Drawing F10523981

CRITICAL	
1. Failure to pass hydrostatic test (as evidenced by absence of stamp)	Visual
2. Failure to pass air test (as evidenced by absence of stamp)	Visual
MAJOR	
101. Concentricity of pitch diameter of nose thread with bourrelets, including perpendicularity of front face of nose	Gage
102. Concentricity of ogive with bourrelets	Gage
103. Concentricity of pitch diameter of rotating disc thread with bourrelets, including perpendicularity of disc seat	Gage
104. Concentricity of body with bourrelets	Gage
105. Concentricity of base with bourrelets (if applicable)	Gage
106. Symmetry of rotating disc seat with major diameter of thread	Gage
107. Diameter of bourrelet(s) min.	Gage
108. Diameter of body	Gage
109. Height of rotating disc seat	Gage
110. Length total, max.	Gage

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4.3.2.4 (Cont'd)

Defect	Method of Inspection
111. Thickness of wall at ogive	Gage
112. Thickness of base through disc seat	Gage
113. Width of rotating disc seat	Gage
114. Improper assembly (nose adapter to body)	Visual (see 6.2)
MINOR	
201. Diameter of rotating disc thread undercut	Gage
202. Height of thread boss	Gage
203. Width of base thread undercut	Gage
204. Depth of depression in base, max.	Visual (see 6.2)
205. Foreign matter	Visual

4.3.2.5 Metal Parts Assembly (After Paint) Drawing F10523981

CRITICAL
None Defined

MAJOR	
101. Pitch diameter of nose thread, max.	Gage
102. Minor diameter of nose thread, max.	Gage
103. Pitch diameter of rotating disc thread, min.	Gage
104. Major diameter of rotating disc thread, min.	Gage
105. Diameter of bourrelets, max.	Gage
MINOR	
201. Pitch diameter of nose thread, min.	Gage
202. Pitch diameter of rotating disc thread, max.	Gage
203. Paint in cavity not dry	Tactile
204. Protective coating inadequate or damaged	Visual
205. Thread(s) damaged or not full	Visual
206. Marking missing, incorrect or illegible	Visual
207. Foreign matter	Visual
208. Protector caps missing	Visual

4.3.2.6 Pallet, Drawing F8848553

CRITICAL
None Defined

MAJOR
None Defined

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4.3.2.6 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Pallet damaged (unserviceable)	Visual
202. Sleeve missing	Visual
203. Improperly packed	Visual
204. Strapping missing, loose or broken	Manual and Visual
205. Marking missing, incorrect or illegible	Visual

4.3.3 Inspection Testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Mechanical Properties

4.3.3.1.1 Tension Test

4.3.3.1.1.1 Body.-Two bodies from each heat treatment, stress relieved, tubing or miscellaneous lot as defined by 4.3.1.2, 4.3.1.3, 4.3.1.4, and 4.3.1.6 as applicable, after final heat treatment or stress relief shall be subjected to this test to determine compliance with the mechanical properties requirements of the applicable drawing. Failure of any tension test specimen to comply shall be cause for rejection of the lot of bodies represented by the failing specimen.

4.3.3.1.1.2 Base.-Two bases from each heat treatment or stress relieved lot as defined by 4.3.1.2, 4.3.1.3 or 4.3.1.6 as applicable shall be subjected to this test to determine compliance with the mechanical properties requirements of the applicable drawing. Failure of any tension test specimen to comply shall be cause for rejection of the lot of bases represented by the failing specimen.

4.3.3.1.1.3 Adapters.-Two adapters shall be selected from each lot and subjected to this test to determine compliance with the mechanical properties requirements of the applicable component drawing. Failure to comply shall be cause for rejection of the lot.

4.3.3.1.1.3.1 Machined from tubing.-With the approval of the procuring activity, when the adapter is machined from tubing in such a manner that the mechanical properties are not altered, test coupons approximately 6 inches in length may be removed from one end of each of 3 tubes to represent the heat lot and tested in lieu of testing the component. Failure of any tension test specimen to comply with the yield strength and elongation requirements of the applicable component drawing shall be cause for rejection of the tubing lot.

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4.3.3.1.1.4 Machined from Bar Stock.-With the approval of the procuring activity when the base or adapter is machined from bar stock in such a manner that the mechanical properties are not altered, test coupons approximately 6 inches in length may be removed from one end of each of three bars representing the heat lot and tested in lieu of testing the component. Failure of any tension test specimens to comply with the yield strength and elongation requirements of the applicable component drawing shall be cause for rejection of the bar stock lot.

4.3.3.2 Brazed Joint

4.3.3.2.1 Hydrostatic Test (A Critical Defect).-Each metal parts assembly shall be subjected to this test to determine compliance with the requirements of the applicable drawing. Failure of any assembly to comply shall be cause for rejection of the assembly.

4.3.3.2.2 Air Pressure Test.-(A Critical Defect) Each metal parts assembly shall be subjected to this test to determine compliance with the requirements of the applicable drawing. Failure of any assembly to comply shall be cause for rejection of the assembly.

4.3.3.2.3 Quality of Joint.-Two body assemblies prior to painting shall be selected from each lot and subjected to this test to determine compliance with 3.4. Failure of any segment of either joint to comply shall be cause for rejection of the lot.

4.3.3.3 Cleaning and Application of Protective Coating.-The sample size and acceptance criteria for the inspection of cleaning the item and the application of protective coating shall be as specified in the applicable specification. Failure to comply with the requirements shall be cause for taking the corrective action recommended therein.

4.3.4 Functional.-The following proving ground tests will be performed by the Government.

4.3.4.1 First Article Sample

4.3.4.1.1 Initial Production Sample.-The quantities listed in Table I shall be ballistically tested in the phase indicated.

TABLE I

Test Phase	Sample Size	
	Initial Production Sample	Production Lot
Excess Pressure	25	10
Service Pressure	15	10

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4.3.4.1.2 Excess Pressure.-Samples provided in accordance with Table I will be subjected to this test. Failure of any metal parts assembly to comply with 3.5.1 shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.3.4.1.3 Service Pressure.-Service pressure test for range velocity will be fired for information. Failure of any metal parts assembly to comply with 3.5.1 shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.3.5 Inspection Equipment.-Inspection Equipment List IEL-10523981 identifies the inspection equipment required to perform the examinations and tests prescribed herein.

4.4 Test Methods and Procedures

4.4.1 Mechanical Properties

4.4.1.1 Tension Test

4.4.1.1.1 Body.-From the bodies selected in accordance with 4.3.3.1.1.1 for the convenience of the Government one tension test specimen conforming to the standard "R" type specimen as prescribed by Federal Test Method Standard No. 151 shall be taken longitudinally from each side of each body 180 degrees apart. The criteria for determining yield strength and elongation shall be as prescribed by Federal Test Method Standard No. 151. Yield strength shall be determined by the offset method. The limiting set shall be 0.1 percent (0.001 inch per inch of gage length.)

4.4.1.2 Base.-From each base selected in accordance with 4.3.3.1.1.2 one tension test specimen conforming to largest obtainable standard "R" type specimen as prescribed by Federal Test Method Standard No. 151 shall be taken from the center of base in a transverse direction normal to the centerline of the part. The criteria for determining yield strength and elongation shall be as prescribed by Federal Test Method Standard No. 151. Yield strength shall be determined by the offset method, the limiting set shall be 0.1 percent (0.001 inch per inch of gage length).

4.4.1.3 Adapters.-From each adapter selected in accordance with 4.3.3.1.1.3, one tension test specimen conforming to the largest obtainable "R" type specimen as prescribed by Federal Test Method Standard No. 151 shall be taken longitudinally from the adapter, and tested in accordance with the criteria for determining yield strength and elongation as prescribed by Federal Test Method No. 151. Yield strength shall be determined by the offset method. The limiting set shall be 0.1 percent (0.001 inch per inch of gage length).

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4.4.1.3.1 Machined from Tubing.-From the test coupons supplied in accordance with 4.3.3.1.1.3.1 one tension test specimen conforming to the largest obtainable standare "R" type tension test specimen as prescribed by Federal Test Method Standard No. 151 shall be taken longitudinally from each test coupon and tested in accordance with the criteria for determining yield strength and elongation as prescribed by Federal Test Method No. 151. Yield strength shall be determined by the offset method. The limiting set shall be 0.1 percent (0.001 inch per inch of gage length).

4.4.1.4 Machined from Bar Stock.-From the samples by 4.3.3.1.1.4 one type "R" tension test specimen shall be taken longitudinally from the center of one end of each bar. The criteria for determining yield strength and elongation as prescribed by Federal Test Method Standard No. 151. Yield strength shall be determined by the offset method, the limiting set shall be 0.1 percent (0.001 per inch per inch of gage length).

4.4.2 Brazed Joint

4.4.2.1 Hydrostatic Pressure.-This test will be performed using equipment capable of performing the test accurately and consistently. In performing the test, the pressure will be applied gradually until the required minimum pressure has been reached, after which the pressure will be maintained for the minimum prescribed time. The equipment shall be calibrated prior to the start of each day's operation, and at intervals of not more than 4 hours of operation. The equipment shall also be calibrated prior to restarting the operation after a shut down of one hour or more. After acceptance stamp letter "H" as indicated on assembly drawing.

4.4.2.2 Air Pressure.-This test will be performed (after the hydrostatic test and drying as specified by the applicable drawing) using equipment capable of performing the test accurately and consistently. The equipment shall be calibrated prior to the start of each day's operation and after each 4 hours continuous operation. The assembly shall be submerged in water in such manner that the base and nose adapter joints are covered. Observation shall be made for air bubbles rising to the surface of the water or bubbles leaving the vicinity of the base or adapter joints. In performing the test the pressure will be applied until the required minimum pressure is reached, after which the pressure will be held for the minimum length of time specified on the applicable drawing. After acceptance stamp letter "A" as indicated on the assembly drawing.

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4.4.2.3 Sectionalized Specimens.-The projectile assemblies selected in accordance with 4.3.3.2.3 shall be sectionalized by cutting through the body at least two inches to the rear of the brazed area at the forward joint and at least two inches to the front of the brazed area of the rear joint. Each of the sections containing the brazed joints shall be cut into two 180 degree segments.

4.4.2.3.1 Quality of Joint.-The sidewall shall be peeled or stripped from each segment by applying continuous pressure. The braze surface of each segment of the base (if applicable) and the adapter and each segment of the respective side wall shall be visually examined to determine compliance with the requirements of 3.4.

4.4.3 Protective Coating.-The procedures for testing the cleaning of components and assemblies and the application of protective coating shall be in accordance with the applicable drawings and the applicable specifications listed thereon. The salt spray test shall be performed utilizing the equipment and procedures prescribed by Federal Test Method Standard No. 141.

4.4.4 Functional

4.4.4.1 Excess Pressure.-The metal parts assemblies provided by Table I for this phase, inert loaded to the service weight will be assembled into complete rounds and fired in a gun for which the round is standard. The propellant charge will be adjusted to obtain 112 percent of the standard service pressure of the round in a new gun. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile on the initial production sample and on the first 5 production lot samples. An attempt will be made to recover at least 60 percent of the projectiles fired in this phase. The recovered projectile will be examined to determine compliance with 3.7.1.

4.4.4.2 Service Pressure.-The metal parts assemblies provided by Table I for this phase, inert loaded to the service weight, will be assembled into complete rounds utilizing the service charge and fired in a gun for which the round is standard. Test rounds will be fired alternately with a similar group of reference rounds for comparison of range and velocity. Velocity and chamber pressure will be recorded for each round fired.

5. PREPARATION FOR DELIVERY

5.1 Packing

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5.1.1 Level - C.-Metal parts assemblies shall be palletized in accordance with drawing F8848553 and MIL-STD-1169.

5.2 Data Card.-Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Functional.-Rounds fired at service pressure for range-velocity are not acceptance tests but results will be carefully analyzed by the Munitions Command to determine quality of the ammunition produced. Evidence of excessive variation in any phase of the functional tests will result in immediate investigation by the Munitions Command.

6.2 Visual Examination Qualification.-When compliance with the applicable requirements are in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

Custodian:

Army - MU

Preparing activity

Army - MU

Project No. 1315 - A622

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